

Introduction to Microeconomics
Lecture 8—Consumer Theory: Preferences

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A model of consumer Preferences

- We start with a set of objects (bundles) X . Elements from from this set, like x or y , are the choices that the consumer can make.
- Most applications consider $X \subset \mathbb{R}_+^n$.
- In this class, $X = \mathbb{R}_+^2$. A bundle x is then $x = (x_1, x_2)$. Interpret it as: x_1 is the quantity of good 1 and x_2 is the quantity of good 2.

Consumer preferences: The Questionnaire Q

Consists of a long list of questions of the following form.

$Q(x, y)$, x and y are different bundles. How do you compare x to y ? Check one and only one of the following three options:

- ☐ I strictly prefer x to y ($x \succ y$).
- ☐ I strictly prefer y to x ($y \succ x$).
- ☐ I am indifferent between x and y ($x \sim y$).

A model of consumer preferences is just a long list of questions of this type and their answers!

What is a legal answer to the questionnaire?

- We need to put some restrictions (assumptions) on answers.

Restriction 1: Completeness

For any question $Q(x, y)$ exactly one of the boxes is checked. We don't allow

- Refraining to answer (no box checked).
- Checking more than one box.
- This means we exclude attitudes like:
 - They are not comparable.
 - I don't know what x is.
 - I have no opinion.
 - I prefer x over y and y over x .
- We are also not allowing attitudes like:
 - It depends on what others think.
 - It depends on the circumstances (frame).
 - Any consideration that includes **intensity**, like I somewhat prefer x , or I love x and hate y .

Restriction 2: Transitivity

Suppose I have bundles x, y, z such that $Q(x, y) : x \succ y$, and $Q(y, z) : y \succ z$. Then this implies that your answer to $Q(x, z)$ should be $x \succ z$.

Similarly, if $Q(x, y)$ is $x \sim y$ and $Q(y, z)$ is $y \sim z$ then in $Q(x, z)$ your answer should be $x \sim z$.

$$\text{Whenever } \left. \begin{array}{l} x \succ y \\ y \succ z \end{array} \right\} \Rightarrow x \succ z$$

$$\left. \begin{array}{l} x \sim y \\ y \sim z \end{array} \right\} \Rightarrow x \sim z$$

- Questionnaire Q could have been rephrased instead with the following type of questions:
 - $Q(x, y)$, x and y are different bundles. How do you compare x to y ? Check one and only one of the following three options:
 - ☐ I consider x at least as good as y ($x \succsim y$).
 - ☐ I consider y at least as good as x ($y \succsim x$).
- What is the relationship between the answers to questionnaire Q and the one above?
 - If $x \succsim y$ and $y \succsim x$ then $x \sim y$.
 - If $x \succsim y$ but $y \not\succsim x$ then $x \succ y$.
 - If $y \succsim x$ but $x \not\succsim y$ then $y \succ x$.

$$\mathbb{R}_+^2 \quad x \succsim y \Leftrightarrow (x_1, x_2) \succsim (y_1, y_2) \Leftrightarrow \begin{matrix} x_1 \succsim y_1 \\ x_2 \succsim y_2 \end{matrix}$$

Suppose the set of bundles $X \subset \mathbb{R}_+^n$ consists of *goods that are desirable*.

Restriction 3: Monotonicity or “more-is-better”

If $x \succsim y$, where $\succsim$ denotes the component-wise standard order, then $x \succsim y$.

If $x > y$, where $>$ denotes the component-wise strict order, then $x \succ y$.

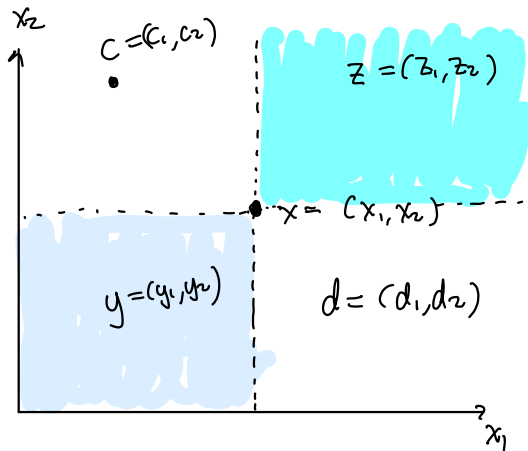
Example: say good 1 is food and good 2 is clothing. $x = (30, 40)$ and $y = (20, 30)$. If preferences satisfy monotonicity, then the answer to $Q(x, y)$ should be $x \succ y$.

- Any relation R over X is called a **preference relation** if it consists of a set of legal answers to the questionnaire Q , that is, if it satisfies completeness and transitivity.

$$\mathbb{R}_+^2 \quad x > y \Leftrightarrow \begin{matrix} x_1 \succ y_1 \\ x_2 \succ y_2 \end{matrix} \quad \text{and the inequality is strict for at least one.}$$

Graphical representations

$X = \mathbb{R}_+^2$ (Assumptions) Restrictions 1 - 3



$z \succ x$

$z_1 > x_1$ AND $z_2 > x_2$

$x \succ y$

$x_1 > y_1$ AND $x_2 > y_2$

Graphical representations: Indifference curves

- For a bundle x the **indifference curve at x** is the set of all bundles that are indifferent to x :

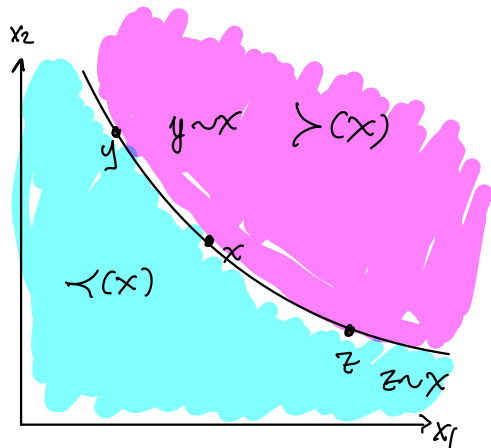
$$IC(x) = \{y \in X : y \sim x\}$$

- The **strictly preferred set to x** is the set of all bundles that are strictly preferred to x :

$$\succ(x) = \{y \in X : y \succ x\}$$

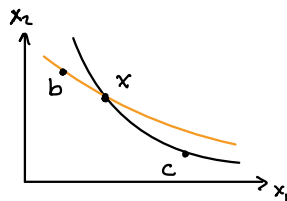
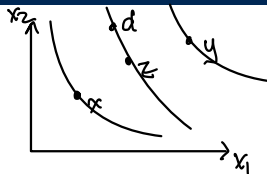
- The **strictly worse set to x** is the set of all bundles that are worse to x :

$$\prec(x) = \{y \in X : x \succ y\}$$

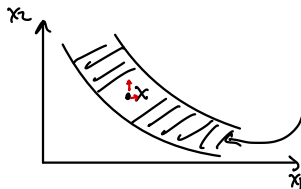


Implications of the restrictions

- Completeness implies that the indifference curve map is **full**, through each bundle there is an indifference curve passing through it.
- Transitivity implies that the indifference curves **cannot intersect**!
- Monotonicity implies that the indifference curves **are not thick**!



$$\begin{aligned}
 &b \sim x && b \text{ vs } c \\
 &x \sim c && b \succ c \\
 &&& (b \succ c, \underbrace{c \sim x \sim b}_{c \sim b})! \\
 &c \succ \underbrace{b \sim x \sim c}_{c \succ c}!
 \end{aligned}$$



~~$I(x)$~~

From Preferences to Utility

- Although preference relations provide a model of the consumers preferences, they are somewhat cumbersome to work with.
- Recall: a preference relation provides an **ordering** over the consumption bundles.
- Idea: can we get some mathematical object that is more tractable to work with, while still preserving the idea that a preference relation provides an ordering over consumption bundles?
- The answer is **Yes!** $(x_1, x_2) \mapsto u(x_1, x_2)$
- We say that a **utility function** $u : X \rightarrow \mathbb{R}$ represents the preference relation $\succsim$ over X if for every $x, y \in X$, $x \succsim y \Leftrightarrow u(x) \geq u(y)$.
- Given a preference relation, can we always ensure that a utility function exists?
 - If $\succsim$ is complete and transitive and X is finite or (countable), the answer is Yes.
 - On the other hand, if X is infinite, like $\mathbb{R}_+^2$, then we need an extra technical restriction: **continuity**, but this will be discussed in higher level courses.

Utility is an ordinal concept, not a cardinal one.

- Suppose that a preference relation $\succsim$ can be represented by the utility function u .
- Do the numbers $u(x)$ for each $x \in X$ matter?
- The answer is no, what matters is the order (Ranking), not the number!
- Utility is ordinal, not cardinal (the magnitude matters).
- The previous idea can be formally stated with the concept of a **monotonic transformation**:
 - Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be an increasing function, i.e. if $a \geq b$ then $f(a) \geq f(b)$.
 - Let u be a utility function representing $\succsim$. Then $v(x) = f(u(x))$ represents the same preferences as u .
 - $x \succsim y \Leftrightarrow u(x) \geq u(y) \Leftrightarrow v(x) \geq v(y)$.

$$\begin{aligned} u(x_1, x_2) = x_1 x_2 &\Leftrightarrow v(x_1, x_2) = \sqrt{x_1 x_2} \Leftrightarrow w(x_1, x_2) = \ln(x_1) + \ln(x_2) \\ x_1 x_2 \geq y_1 y_2 &\Leftrightarrow \sqrt{x_1 x_2} \geq \sqrt{y_1 y_2} \Leftrightarrow \ln(x_1) + \ln(x_2) \geq \ln(y_1) + \ln(y_2) \end{aligned}$$

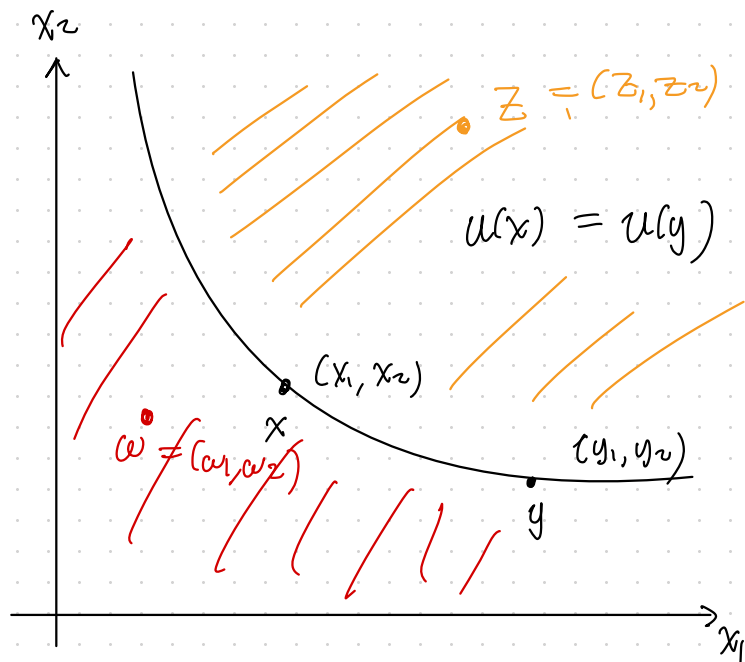
- With the help of a utility function, it becomes easy to define an indifference curve:

$$x \sim y \Leftrightarrow u(x) = u(y)$$

- Then the indifference curve passing through x is the set of consumption bundles that achieve the same level of utility $u(x)$:

$$IC(x) = \{y \in X : u(y) = u(x)\}$$

- We can define in a similar way the strictly preferred and the strictly worse sets.



$u(z_1, z_2) > u(x_1, x_2)$
 (strictly preferred set / upper contour)

$u(w_1, w_2) < u(x_1, x_2)$
 (worse than set / lower contour)

$u(x_1, x_2)$

Example: $X = \mathbb{R}_+^2$

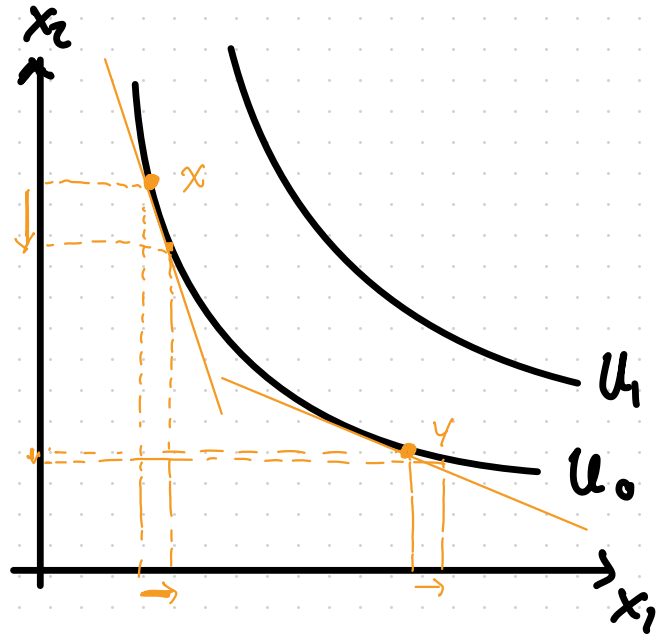
$$u(x_1, x_2) = x_1 x_2$$

Fix a level of utility $u_0 = x_1 x_2$

$\hookrightarrow$
solve
for x_2

$$x_2 = \frac{u_0}{x_1}$$

$$u_1 > u_0$$



The Marginal Rate of Substitution

- It is the absolute value of the slope of the indifference curve at a particular point.
- It measures the cost of opportunity of good 1 in terms of good 2 while staying on the same indifference curve.
- Intuitively, it is the personal opportunity cost (in terms of the consumer's preferences) of remaining in the same indifference curve, measured as units of good 2 that must be given up to get one more unit of good 1.
- How? Let's compute the total change in utility, when the utility function is $u(x_1, x_2)$:

$$\Delta U = \frac{\Delta U}{\Delta x_1} \Delta x_1 + \frac{\Delta U}{\Delta x_2} \Delta x_2$$

MU_1 $\swarrow$ $\searrow$ MU_2

$$\Delta U = MU_1 \cdot \Delta x_1 + MU_2 \cdot \Delta x_2$$

- At an indifference curve, utility is not changing, so $\Delta U = 0$, this leads to:

$$\text{slope of IC} = \frac{\Delta x_2}{\Delta x_1} = - \frac{\frac{\Delta U}{\Delta x_1}}{\frac{\Delta U}{\Delta x_2}} = - \frac{MU_1}{MU_2}$$

- The **Marginal Rate of Substitution** is the absolute value of the quantity above.

Example: $X = \mathbb{R}_+^2$

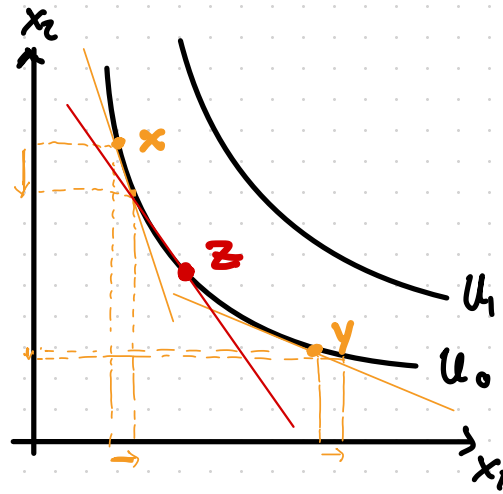
$$u(x_1, x_2) = x_1 x_2$$

Fix a level of utility $u_0 = x_1 x_2$

$\hookrightarrow$ solve for x_2

$$x_2 = \frac{u_0}{x_1}$$

$u_1 > u_0$



$$MU_1 = x_2$$

$$MU_2 = x_1$$

$$\frac{MU_1}{MU_2} = MRS = \frac{x_2}{x_1}$$

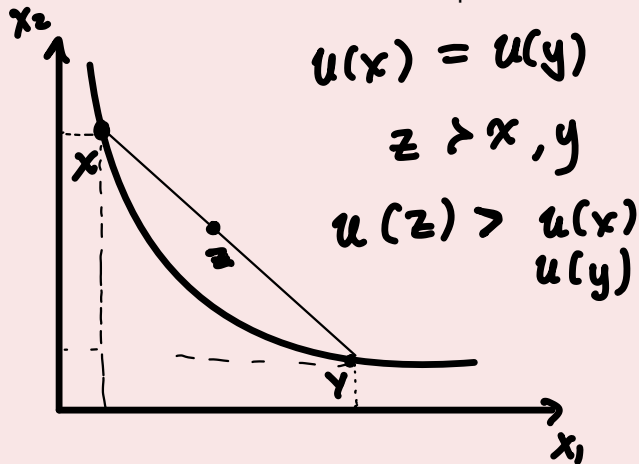
$$\text{slope of IC at } x = (x_1, x_2) = -\frac{x_2}{x_1}$$



In general, the MRS changes with the bundle along the IC
we are considering

Convexity

Preferences that are convex exhibit a preference for more balanced consumption bundles.



- IC are convex for convex preferences
- MRS that is decreasing along the IC.

Examples:

$$U(x_1, x_2) = \sqrt{x_1} + \sqrt{x_2}$$

$$U_0 = \sqrt{x_1} + \sqrt{x_2}$$

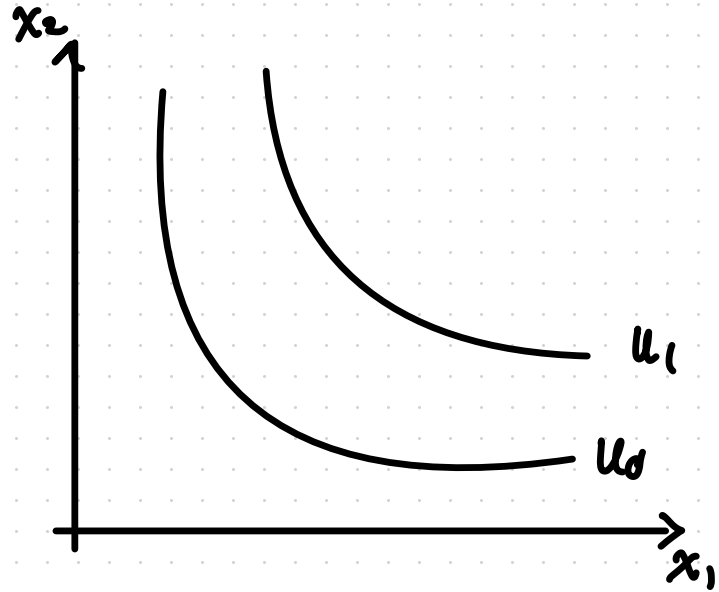
↳
Solve for x_2

$$x_2 = (U_0 - \sqrt{x_1})^2$$

$$MU_1 = \frac{1}{2\sqrt{x_1}}$$

$$MRS = \frac{MU_1}{MU_2} = \frac{\sqrt{x_2}}{\sqrt{x_1}}$$

$$MU_2 = \frac{1}{2\sqrt{x_2}}$$



Perfect Substitutes

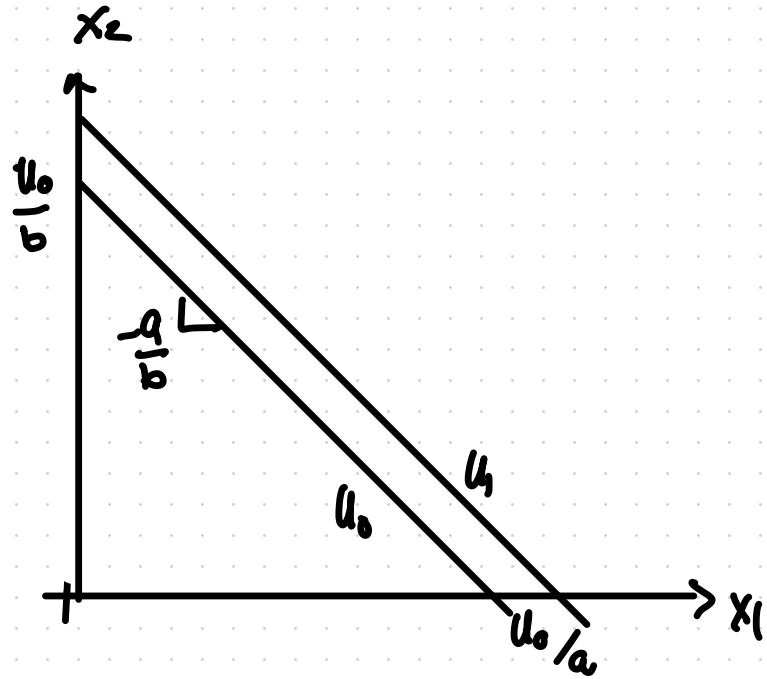
$$U(x_1, x_2) = a x_1 + b x_2$$

$$a, b > 0$$

$$U_0 = a x_1 + b x_2$$

$$x_2 = \frac{U_0}{b} - \frac{a}{b} x_1$$

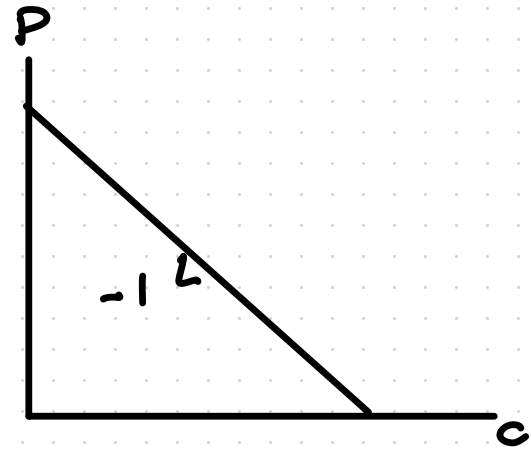
$$MRS = \frac{a}{b} = \text{constant}$$



Prbl. A preferences for C, P

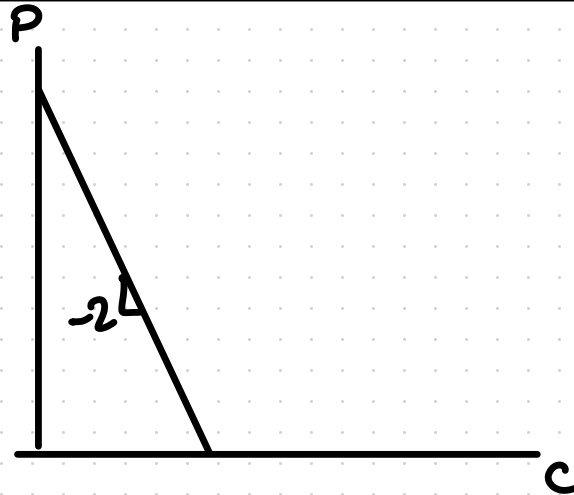
$$u(C, P) = C + P$$

$$MRS = 1$$



$$u(C, P) = 2C + P$$

$$MRS = 2$$



Perfect Complements

Ex: Right shore } Not for
a } pirates
Left shore }

Proportion 1:1

Espresso: Water & Coffee
2 : 1

$$u(x_1, x_2) = \min \left\{ \frac{x_1}{a}, \frac{x_2}{b} \right\}$$

$$a, b > 0$$

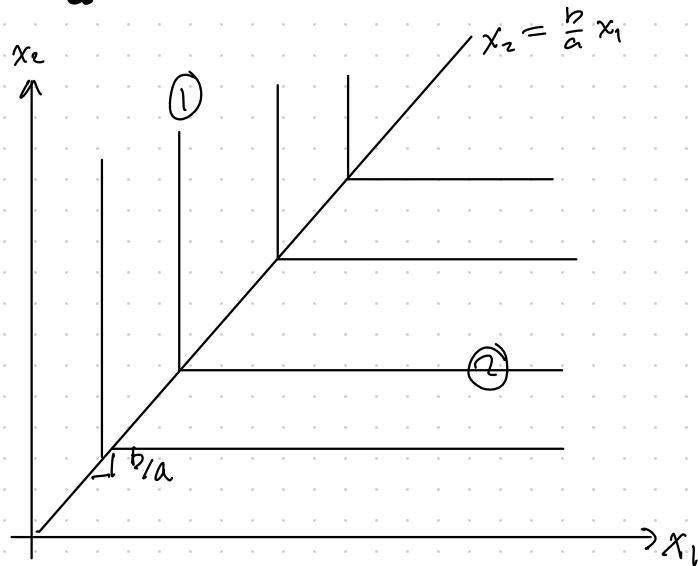
$$\textcircled{1} \quad u_0 = \min \left\{ \frac{x_1}{a}, \frac{x_2}{b} \right\} = \frac{x_1}{a} \Rightarrow u_0 = \frac{x_1}{a}$$

$$\textcircled{2} \quad u_0 = \frac{x_2}{b} \Rightarrow u_0 b = x_2$$

$$u_0 = \min \left\{ \frac{x_1}{a}, \frac{x_2}{b} \right\} \quad \left| \begin{array}{l} \frac{x_1}{a} = \frac{x_2}{b} \Leftrightarrow \\ b/a \cdot x_1 = x_2 \end{array} \right.$$

$$\frac{x_1}{a} < \frac{x_2}{b} \Leftrightarrow \frac{b}{a} x_1 < x_2 \quad (1)$$

$$\frac{x_1}{a} > \frac{x_2}{b} \Leftrightarrow \frac{b}{a} x_1 > x_2 \quad (2)$$



$x_1 = \text{Water (grams)}$ $\frac{2}{1}$

$x_2 = \text{coffee (grams)}$ 1

$$x_2 = \left(\frac{1}{2}\right) x_1$$

$\searrow \frac{b}{a}$

$$u(x_1, x_2) = \min \left\{ \frac{x_1}{2}, x_2 \right\}$$

